Statement sheet: does the frozen Lean statement say what Balaban's printed lemma says?

Object. The Lean proposition `AdalyFormal.StatementLaneG.P428Target8` (proved as `AdalyFormal.M2RightInv.p428Target8`) against the sentence on p. 428 of Balaban, *Propagators for lattice gauge theories in a background field*, CMP 99 (1985) (CMP99b below): "a positive definite operator $C^*\Delta_k C$ with a lower bound $\gamma_0 > 0$ independent of k and $U\dots$ for an arbitrary configuration U satisfying (3.35), (3.36) with $M\alpha_0$ sufficiently small". This sheet is meant to be read alone. Part A has one block per clause (print, Lean, plain words, deviation, a yes/no question). Part B writes the whole statement in ordinary notation with the quantifier order. Part C lists the readings a reader might reject, what changes, and which proved variant covers the alternative.

Conventions. *Print locations.* "p. 396, l. 6–10" counts text lines of the printed page from the first line under the running head, displays not counted; for other pages the corpus transcription is cited as `pN:lM` = line M of `corpus/pages/CMP99b_pN.md` (CMP96 = `CMP96_pN.md`); quotes are verbatim, mathematics in TEX. *Lean locations* are `File.lean:first-last` of the excerpt; full paths and file hashes are in the appendix and in `CHECKS.md`. *Deviation ids* (D1–D12, N1–N6, $\sqrt{2}$) are the rows of the deviation table of the paper, §3.7. In Lean, a field of "extent" L_μ means $L_\mu + 1$ sites; ℓ is Balaban's L.

Part A. Clause by clause

A1 Lattice, torus, scale, distance

Print. CMP96 p.223 (`CMP96_p223:162`): "a distance between two points $x, y \in \eta Z^d$, $\eta > 0$, is given by the l^1-norm of the vector $x - y$ and is denoted by $|x - y|$, i.e. $|x - y| = \sum_{\mu=1}^{d} |x_\mu - y_\mu|$". Domains: $\Omega_j \subset T_\eta$ (p.393, CMP96 p.224 (2.1)). Unit lattice: $\Lambda \subset \Lambda_k = \Omega_k^{(k)}$ (p.427, `p427:139`).

The print does not say the lattice is periodic, nor its dimension in this clause, nor the value of η.

ZStatementP428.lean:42–43,58–59,66–67,286–287

```
def scaleUp (ell k : ℕ) (Lu : PeriodicLatticeExtent) : PeriodicLatticeExtent :=
  fun μ => (Lu μ + 1) * ell ^ k - 1
def pcd (N a b : ℕ) : ℕ :=
  min (if a ≤ b then b - a else a - b) (N - (if a ≤ b then b - a else a - b))
def pdist {L : PeriodicLatticeExtent} (x y : PeriodicLatticeSite L) : ℕ :=
  ∑ μ : Fin 3, pcd (L μ + 1) (x μ).val (y μ).val
def TorusOK (ell M : ℕ) (Lu : PeriodicLatticeExtent) : Prop :=
  (∀ μ : Fin 3, ell ∣ Lu μ + 1) ∧ (∀ μ : Fin 3, ell < Lu μ + 1) ∧ (∀ μ : Fin 3, M ∣ Lu μ + 1)
```

In words. The unit torus has $\mathrm{Lu}\,\mu + 1$ sites per direction; the fine torus (carrying U and the tower) has ℓ^k times as many, so the fine spacing is $\eta = \ell^{-k}$ (it appears below as `((ell:ℝ)^k)⁻¹`). Distance is the sum over the three axes of the shorter way round the circle. `TorusOK`: ℓ and M divide every unit side, and $\ell <$ side.

Deviation. Dimension 3, group SU(2), periodic torus, $\eta = \ell^{-k}$ and the wrap-around in the distance are conventions not printed (paper §3.1, row N4). `TorusOK`: row D7 (fewer tori; blocking hygiene).

Check (yes/no). Is it acceptable, for this lemma, to work on a 3-torus with $\eta = \ell^{-k}$ and periodic l^1 distance, for tori whose sides are multiples of ℓ and M?

A2 The tower $\Omega_0 \supset \cdots \supset \Omega_k$ (`Admissible`)

Print. CMP96 p.224 (`CMP96_p224:138-54`): "We consider a sequence of domains $\Omega_1 \supset \Omega_2 \supset \ldots \supset \Omega_k$, $\Omega_j \subset T_\eta \dots\ \Omega_j = B^j(\Omega_j^{(j)})$, $\Omega_j^{(j)} \subset T^{(j)}_{L^j\eta}$ and it is a sum of big blocks, $(L^j\eta)^{-1}\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) > RM$, M is a size of big blocks and R is a big positive integer which will be fixed later." CMP99b p.393 (`p393:122-40`): "the sequence (2.1) starts with Ω_0, thus we have $\Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k \dots\ \Omega_{k+1} = \varnothing \dots$ The condition (2.2) is unchanged." Big blocks: p.408 (`p408:129`): "elementary cubes (cells) of the lattice $ML^j\eta Z^d$".

ZStatementP428.lean:51–51,54–55,73–82

```
def bigSide (M ell j : ℕ) : ℕ := M * ell ^ j
def BlockClosed {L : PeriodicLatticeExtent} (s : ℕ) (S : Set (PeriodicLatticeSite L)) : Prop :=
  ∀ x ∈ S, ∀ y : PeriodicLatticeSite L, (∀ μ : Fin 3, (y μ).val / s = (x μ).val / s) → y ∈ S
def Admissible {L : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower L) : Prop :=
  (∀ j : ℕ, S.Om (j + 1) ⊆ S.Om j) ∧
  (∀ j : ℕ, k < j → S.Om j = ∅) ∧
  (∀ j : ℕ, j ≤ k → ∀ μ : Fin 3, bigSide M ell j ∣ (L μ + 1)) ∧
  (∀ j : ℕ, j ≤ k → BlockClosed (bigSide M ell j) (S.Om j)) ∧
  (∀ j : ℕ, j ≤ k → ∀ x : PeriodicLatticeSite L, x ∉ S.Om j →
     ∀ y ∈ S.Om (j + 1), R * bigSide M ell j < pdist x y) ∧
  (∀ x : PeriodicLatticeSite L, x ∈ S.Om 0 ↔
     ∃ y ∈ S.Om 1, ∃ z : PeriodicLatticeSite L,
       (∀ μ : Fin 3, (z μ).val / M = (x μ).val / M) ∧ pdist z y ≤ R * M)
```

In words. Clauses in order: (1) nested; (2) empty above level k; (3) a level-j block (side $M\ell^j$ fine sites) tiles each side; (4) each Ω_j is a union of such blocks; (5) a point outside Ω_j is at distance $> RM\ell^j$ (fine sites; equivalently $> RM$ in units of ℓ^j) from Ω_{j+1}; (6) Ω_0 is the union of the M-blocks having a site within RM of Ω_1.

Deviation. Clause (6) is the p.396 choice of Ω_0 (A3). Clauses (3), tiling and padding above k are not printed (N4); ℓ^1 periodic distance (N4); block side read as $M\ell^j$ fine sites (p.408). Row N3/N4: identical to the printed conditions under reading A of T_1 (Lean `RBridge.admissible_iff_print`, `RB1.lean:185`); see Part C.

Check (yes/no). (a) Are big blocks at level j cubes of side $M\ell^j$ fine sites? (b) Is the collar of (2.2) strict ($>$) in fine sites $R \cdot M\ell^j$?

A3 The innermost domain Ω_0 and Λ_0 (last clause of `Admissible`)

Print. CMP99b p.396, l.10–12: "The set Λ_0 in the above condition means $\Lambda_0 = \Omega_0 \backslash \Omega_1$, where $\Omega_0 = \{$a union of big blocks of the lattice T_1, such that their distances to Ω_1 are $\leq RM\}$." p.394 (`p394:174-77`): "Let us introduce a domain Ω_0 such that $\Omega_1 \subset \Omega_0$ and Ω_0 is a union of big blocks of the lattice T_1, e.g. we may take $\Omega_0 = \{$a union of big blocks in T_1, with distances to $\Omega_1 \leqq RM\}$, or we may add to Ω_1 a thinner layer of the big blocks surrounding Ω_1."

In words. Lean (the last conjunct of the excerpt in A2): Ω_0 is exactly the set of M-blocks at distance $\leq RM$ from Ω_1; $\Lambda_0 = \Omega_0 \setminus \Omega_1$ is the stratum of index 0 (A5).

Deviation. Two readings are made: T_1 is the fine lattice in its own unit spacing ("reading A"; the alternative "reading B" takes T_1 as the level-k unit lattice); and p.394 allows a thinner layer, while Lean fixes the $\leq RM$ layer. Rows N3, N4; Part C items 1–2.

Check (yes/no). Is "T_1" in the sentence on p.396 the fine lattice at spacing one fine site (blocks of M fine sites, radius RM fine sites)?

A4 The region Λ (`LambdaOK2`)

Print. CMP99b p.427 (`p427:139`): "We assume that $\Lambda \subset \Lambda_k = \Omega_k^{(k)}$, Λ is a union of big blocks, and a distance between Λ and Λ_k^c is bigger than RM." CMP96 p.245, Lemma 2.4 (`CMP96_p245:195`): "Let a set $\Lambda \subset Z^d$ be a sum of blocks, $\Lambda = B(\Lambda')$."

ZStatementP428.lean:189–195,203–205; ZT4A.lean:57–58,88–90

```
def LambdaOK {Lu Lf : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower Lf)
    (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  BlockClosed M Lam ∧
  ∃ emb : PeriodicLatticeSite Lu → PeriodicLatticeSite Lf,
    (∀ y μ, (emb y μ).val = (ell ^ k) * (y μ).val) ∧
    (∀ y ∈ Lam, emb y ∈ S.Om k) ∧
    (∀ y ∈ Lam, ∀ y' : PeriodicLatticeSite Lu, emb y' ∉ S.Om k → R * M < pdist y y')
def LamPrime {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu)) :
    Set (PeriodicLatticeSite Lu) :=
  {y | y ∈ CoarseLattice Lu ell ∧ ∀ x ∈ blockOf y ell, x ∈ Lam}
def LamBlockSum {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  ∀ x ∈ Lam, ∃ y ∈ LamPrime ell Lam, x ∈ blockOf y ell
def LambdaOK2 {Lu Lf : PeriodicLatticeExtent} (ell M R k : ℕ) (S : Tower Lf)
    (Lam : Set (PeriodicLatticeSite Lu)) : Prop :=
  LambdaOK ell M R k S Lam ∧ LamBlockSum ell Lam
```

In words. Λ is a set of unit-torus sites, a union of M-blocks; scaling a unit site y to the fine site $\ell^k y$ lands in Ω_k; every unit site y' whose image is not in Ω_k is farther than RM (unit sites) from every site of Λ; and Λ is the union of the ℓ-blocks lying inside it ($\Lambda = B(\Lambda')$). The empty Λ is allowed.

Deviation. Row N2: equal to the print; $\Lambda = \varnothing$ allowed (inequality then trivial, `OneGap.instIneq_empty`). The collar uses the same R as the tower; this is matched by reading, not by a theorem (paper §3.5).

Check (yes/no). Is "RM in the unit lattice" the same R as in the tower condition (2.2)?

A5 The class of cubes (`IsClassCube`)

Print. CMP99b p.396, l.6–10: "For each cube $\square$ of this class there exists a unique index j, $0 \le j \le k$, such that $\square \subset B^j(\Lambda_j) \cup B^{j+1}(\Lambda_{j+1})$, $\square \cap B^j(\Lambda_j) \neq \emptyset$, and $\square$ is a union of several big blocks of the lattice $T_{L^{-j}}$, which implies that its size in the lattice T_η is $O(1)ML^j\eta$. Here $O(1)$ will mean a number ≥ 10." p.408/409 (`p408:131`, `p409:120`): "We have assumed that the number $O(1)$ in the condition (3.35) can be taken as equal to 12".

ZStatementP428.lean:131–137,140–142

```
def DescribedAt {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (sites : Set (PeriodicLatticeSite L)) (i : ℕ) : Prop :=
  i ≤ k ∧
  (∃ z : PeriodicLatticeSite L, GridCorner (bigSide M ell i) z ∧
     sites = siteBox z (f * bigSide M ell i)) ∧
  sites ⊆ towerBj S i ∪ towerBj S (i + 1) ∧
  (sites ∩ towerBj S i).Nonempty
def IsClassCube {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (sites : Set (PeriodicLatticeSite L)) (j : ℕ) : Prop :=
  DescribedAt ell M f k S sites j ∧ ∀ i : ℕ, DescribedAt ell M f k S sites i → i = j
```

In words. With a cube factor f: a class cube of index i is the box of side $f \cdot M\ell^i$ fine sites whose corner has coordinates divisible by $M\ell^i$, lying in $(\Omega_i \setminus \Omega_{i+1}) \cup (\Omega_{i+1} \setminus \Omega_{i+2})$, meeting $\Omega_i \setminus \Omega_{i+1}$, with i the only index for which this holds. (Here $B^j(\Lambda_j) = \Omega_j \setminus \Omega_{j+1}$, p.393.) The headline uses the single number $f = 12$.

Deviation. D3: the print attaches $O(1) \ge 10$ to each cube and says "union of several big blocks"; Lean takes one number for all cubes and boxes of f blocks per side. Proved variants with a free factor, per-cube factors, and unions of blocks: Part C, items 3–4. The scale "$T_{L^{-j}}$" is read as blocks of $M\ell^j$ fine sites.

Check (yes/no). Is fixing one cube factor (12, as p.409 says) and taking cubes to be boxes an acceptable reading of the p.396 class for this lemma?

A6 The regularity class (3.35), (3.36) (`PerCubeRegular2`)

Print. CMP99b p.396 (`p396:135-48`): "for an arbitrary cube $\square$ of the described above class, and for a configuration U there exists a gauge transformation u on $\square$ such that $U^u = e^{i\eta A}$, and if the index of $\square$ is j, then $|A| < O(1)M\alpha_0(L^j\eta)^{-1}$, $|\nabla^\eta A| < O(1)M\alpha_0(L^j\eta)^{-2}$ on $\square$, where $O(1)M$ is a size of $\square$ in $T_{L^{-j}}$; $|\partial^{\eta *}\partial^\eta A| < O(1)M\alpha_0(L^j\eta)^{-3}$ on $\square$." (3.35), (3.36). Bonds of a set: CMP96 p.224 (`CMP96_p224:161`): "bonds $b \subset T_\eta$: at least one end-point of b belongs to Ω".

ZStatementP428.lean:145–146; ZR2A.lean:47–66,29–32,40–43

```
def CubeBond {L : PeriodicLatticeExtent} (sites : Set (PeriodicLatticeSite L)) (b : Bond L) : Prop :=
  b.1 ∈ sites ∧ periodicLatticeShift b.1 b.2 ∈ sites
def PerCubeRegular2 {L : PeriodicLatticeExtent} (ell M f k : ℕ) (S : Tower L)
    (α₀ η : ℝ) (U : LinkCfg L) : Prop :=
  ∀ (sites : Set (PeriodicLatticeSite L)) (j : ℕ), IsClassCube ell M f k S sites j →
    ∃ (g : PeriodicLatticeSite L → Matrix.specialUnitaryGroup (Fin 2) ℂ)
      (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ),
      (∀ e : Bond L, CubeBond sites e →
        HT (A e) ∧
        ((periodicGaugeTransform g U e : Matrix.specialUnitaryGroup (Fin 2) ℂ) :
            Matrix (Fin 2) (Fin 2) ℂ)
          = NormedSpace.exp (((η : ℂ) * Complex.I) • A e) ∧
        frobNorm (A e) < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^-1 ^ 1)) ∧
      (∀ (μ ν : Fin 3) (x : PeriodicLatticeSite L),
        CubeBond sites ((x, ν) : Bond L) → CubeBond sites ((periodicLatticeShift x μ, ν) : Bond L) →
        frobNorm ((η^-1 : ℝ) • (A (periodicLatticeShift x μ, ν) - A (x, ν)))
          < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^-1 ^ 2)) ∧
      (∀ (μ : Fin 3) (x : PeriodicLatticeSite L),
        (∀ ν : Fin 3, ν ≠ μ → CurlReads sites (periodicLatticeShiftBack x ν) ν μ ∧
          CurlReads sites x ν μ) →
        frobNorm (ddA η A x μ)
          < (f : ℝ) * M * α₀ * (((ell : ℝ) ^ j * η)^-1 ^ 3))
noncomputable def curlA {L : PeriodicLatticeExtent} (η : ℝ) (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ)
    (x : PeriodicLatticeSite L) (a b : Fin 3) : Matrix (Fin 2) (Fin 2) ℂ :=
  (η^-1 : ℝ) • (A (periodicLatticeShift x a, b) - A (x, b)
    - A (periodicLatticeShift x b, a) + A (x, a))
noncomputable def ddA {L : PeriodicLatticeExtent} (η : ℝ) (A : Bond L → Matrix (Fin 2) (Fin 2) ℂ)
    (x : PeriodicLatticeSite L) (μ : Fin 3) : Matrix (Fin 2) (Fin 2) ℂ :=
  (η^-1 : ℝ) • ∑ ν ∈ Finset.univ.filter (fun ν : Fin 3 => ν ≠ μ),
    (curlA η A (periodicLatticeShiftBack x ν) ν μ - curlA η A x ν μ)
```

In words. U is regular with parameter α_0 if for every class cube (index j) *there are* a gauge transformation g and a Hermitian traceless field A with $U^g = \exp(i\eta A)$ on the bonds of the cube, $\|A\| < fM\alpha_0(\ell^j\eta)^{-1}$, the forward difference of A (divided by η) is

$< fM\alpha_0(\ell^j\eta)^{-2}$ where both bonds are in the cube, and $\partial^*\partial A$, written as the lattice curl-of-curl `ddA`, is $< fM\alpha_0(\ell^j\eta)^{-3}$ where all bonds it reads are in the cube. g and A depend on the cube.

Deviation. N6: "$\exists g, A$ per cube" equals the print. $\sqrt{2}$: norm of A is the full-matrix Frobenius norm, $\sqrt{2}$ times the printed norm (absorbed in α_0). D9: $\partial^*\partial$ replaced by the curl form. Bond convention: `CubeBond` needs *both* endpoints in the cube; CMP96 p.224 says *at least one*. Part C, item 5.

Check (yes/no). (a) Is the existential gauge g chosen per cube, inside the "for every cube"? (b) Is "bonds of the cube = both endpoints in the cube" acceptable for the class conditions?

A7 Coverage and the single hypothesis on U (`ClassCoupled`)

Print. CMP99b p.396, l.after the display (p396:151): "Let us notice that these regularity conditions do not impose any constraints on U outside the domain Ω_0."

Coverage is not printed.

ZR2A.lean:107–109; ZR6A.lean:27–30,33–34,124–125

```
def CoversOm0c {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  ∀ x ∈ S.Om 0, ∃ (sites : Set (PeriodicLatticeSite L)) (j : ℕ),
    IsClassCube ell M c k S sites j ∧ x ∈ sites
def CoversPlaqOm0c {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  ∀ p : PeriodicLatticePlaquette L, (∀ e ∈ periodicPlaquetteBoundaryEdges p, CubeBond (S.Om 0) e) →
    ∃ (sites : Set (PeriodicLatticeSite L)) (j : ℕ), IsClassCube ell M c k S sites j ∧
      ∀ e ∈ periodicPlaquetteBoundaryEdges p, CubeBond sites e
def CoversOm0p {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) : Prop :=
  CoversOm0c ell M c k S ∧ CoversPlaqOm0c ell M c k S
def ClassCoupled {L : PeriodicLatticeExtent} (ell M c k : ℕ) (S : Tower L) (α₀ η : ℝ) (U : LinkCfg L) : Prop :=
  CoversOm0p ell M c k S ∧ PerCubeRegular2 ell M c k S α₀ η U
```

In words. Every site of Ω_0 lies in a class cube of factor c; every plaquette whose four boundary bonds lie in Ω_0 has its four bonds in one class cube of factor c; and U is regular on all class cubes with the same c. The hypothesis on U in the lemma is this conjunction.

Deviation. D2: an added hypothesis (the print says the conditions constrain U only inside Ω_0; Lean reads: they constrain U throughout Ω_0). It restricts the towers. Without it the encoded statement is false (Part C, item 6).

Check (yes/no). Is "the class cubes cover Ω_0 (sites and plaquettes)" a fair reading of "the conditions are imposed on Ω_0"?

A8 Site operators: Δ'_a, G', R, W, Δ_π

Print. CMP99b p.394 (p394:160–84): "$\Delta'_a = (\Delta^\eta_U + Q'^* a Q') \restriction_{\Omega_0} \dots \langle \lambda, Q'^* a Q' \lambda\rangle = \sum_{j=0}^{k} a_j \sum_{y\in\Lambda_j} (L^j\eta)^{d-2} |(Q'_j(U)\lambda)(y)|^2$ (3.24), $a_{j+1} = aa_j/(aL^{-2}+a_j)$, $a_1 = a_0 = a > 0 \dots$ $\Delta'_a \restriction_{\Omega_0} = \Omega_0\Delta'_a\Omega_0 \dots$ Its inverse is denoted by G'". (3.21) R is the orthogonal projection onto $\Delta_U N(Q')$ (p.394). (3.119) p.419: $\langle A, \Delta_\pi A\rangle = \langle A - DG'RD^*A, \Delta(A - DG'RD^*A)\rangle$.

R2Ops1.lean:121–125; ZXR2Wire1.lean:20–24; ZR2Ops2.lean:27–30,57–58,63–64; R2Ops1.lean:162–163,170–171; ZR2Ops2.lean:67–68,83–84

```
structure TowerGeom (L : PeriodicLatticeExtent) where
  ell : ℕ
  k : ℕ
  Om0 : Set (PeriodicLatticeSite L)
  Lam : ℕ → Set (PeriodicLatticeSite L)
def geomOf (ell k : ℕ) {Lf : PeriodicLatticeExtent} (S : Tower Lf) : TowerGeom Lf where
  ell := ell
  k := k
  Om0 := S.Om 0
  Lam := fun j => {x | (∀ μ : Fin 3, ell ^ j | (x μ).val) ∧ x ∈ S.Om j ∧ x ∉ S.Om (j + 1)}
noncomputable def Qpa (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  ∑ j ∈ Finset.range (G.k + 1), ∑ y : PeriodicLatticeSite L,
    if y ∈ G.Lam j then (ajSeq a G.ell j * (G.ell : ℝ) ^ j) •
      ((ContinuousLinearMap.adjoint (Qlev G U j y)).comp (Qlev G U j y)) else 0
noncomputable def DelpA (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  ((siteMask G.Om0).comp (lapFull U + Qpa G U a)).comp (siteMask G.Om0)
noncomputable def Gp (G : TowerGeom L) (U : Cfg L) (a : ℝ) : SiteL2 L →L[ℝ] SiteL2 L :=
  (Ring.inverse (DelpA G U a + (1 - siteMask G.Om0))).comp (siteMask G.Om0)
noncomputable def lapDir (G : TowerGeom L) (U : Cfg L) : SiteL2 L →L[ℝ] SiteL2 L :=
  ((siteMask G.Om0).comp (lapFull U)).comp (siteMask G.Om0)
noncomputable def Rproj (G : TowerGeom L) (U : Cfg L) : SiteL2 L →L[ℝ] SiteL2 L :=
  (RImage G U).starProjection
noncomputable def Wop (G : TowerGeom L) (U : Cfg L) (a : ℝ) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  1 - (gradL2 U).comp ((Gp G U a).comp ((Rproj G U).comp (gradAdj U)))
noncomputable def DpiFull (G : TowerGeom L) (U : Cfg L) (a : ℝ) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  (ContinuousLinearMap.adjoint (Wop G U a)).comp ((Delta G U).comp (Wop G U a))
```

In words. The tower geometry (Ω_0 and the strata Λ_j at coarse sites) is read off the tower. `Qpa` is the mass term (3.24); `DelpA` is $\Omega_0(\Delta + Q'^*aQ')\Omega_0$; `Gp` is its inverse on $L^2(\Omega_0)$, extended by zero; R is the projection onto $\Delta_U N(Q')$ with Dirichlet Laplacian (`lapDir`); $W = I - DG'RD^*$; $\Delta_\pi = W^*\Delta W$. `Ring.inverse` is Lean's inverse, with value 0 where there is none.

Deviation. D10, D11: the averaging operators $Q_j(U)$ come from the one-step jets of the print's (3.13)–(3.15), per component; convention, bridged but not re-derived. The Lean body is written at $\eta = 1$ ("η-free"; see A12). Not pinned to a printed page: that the weights $L^{j(d-2)}$ versus `ell^j` in `Qpa` equal the print's in $d = 3$ (paper §3.6 records them as inherited, not re-derived).

Check (yes/no). Do `DelpA`, `Gp`, `Wop`, `DpiFull` correspond to (3.24), the inverse G', W of (3.119), with Dirichlet conditions on Ω_0^c?

A9 The operators Δ, J and the plaquette selection

Print. CMP99b p.392 (p392:187–104): $\langle A, \Delta A\rangle = \langle A, D^*DA\rangle + \langle A, \Delta' A\rangle$, $\langle A, \Delta' A\rangle = \sum_{p\subset T_\eta} \eta^d \mathrm{tr}((D^1_U A)(p))^2 \eta^{-2}(\mathrm{Re}\, U(\partial p) - 1) + \dots$ (3.10); $\langle A, J\rangle = \sum_{b\subset T_\eta} \eta^d \mathrm{tr} A(b) J(b)$, $J = D^*\eta^{-2}\mathrm{Im}\,\partial U$ (3.11). p.394 (p394:181–84): "They depend on the configuration U restricted to $\Omega_0 \dots$ All operators we will consider will be defined by using Dirichlet boundary conditions on Ω_0."

The print sums (3.10), (3.11) over all *plaquettes of T_η.*

ZR2Ops2.lean:74–75,79–80; ZXR2Wire2.lean:51–52

```
def plaqIn (S : Set (PeriodicLatticeSite L)) : Set (PeriodicLatticePlaquette L) :=
  {p | ∀ e ∈ periodicPlaquetteBoundaryEdges p, e.1 ∈ S ∧ periodicLatticeShift e.1 e.2 ∈ S}
noncomputable def Delta (G : TowerGeom L) (U : Cfg L) : BondFnL2 L →L[ℝ] BondFnL2 L :=
  AdalyFormal.StatementLaneR6.deltaKLocCLM U (plaqIn G.Om0)
noncomputable def Jf : BondFnL2 (scaleUp P.ell P.k P.Lu) :=
  WithLp.toLp 2 (fun b => -(AdalyFormal.IngredientFamily.JSU2Loc 1 (plaqIn P.G.Om0) P.U b))
```

In words. Δ and J are summed only over plaquettes whose four boundary bonds have both endpoints in Ω_0, so they read U only on Ω_0. (J carries the sign of the chart used for $\mathfrak{su}(2)$.)

Deviation. D1: a chosen reading of p.394, not the printed sums. Two literal alternatives (all plaquettes; a bond mask inside $W^*\Delta W$) are refuted by Lean theorems (Part C, item 7).

Check (yes/no). Is localising Δ and J to plaquettes inside Ω_0 an acceptable meaning of "the operators depend on U restricted to Ω_0"?

A10 G_1, H_1, $\tilde{D}^{(2)}$, Q, Q_1 (the pieces of (3.156))

Print. CMP99b (3.126) p.420 (p420:182): $HB = GQ^*(QGQ^*)^{-1}B$. (3.128) p.421 (p421:130): $\langle A, G_1^{-1}A\rangle = \langle A, \Delta_\pi A\rangle - 2\langle HC_\pi^{(2)}(A), J\rangle + \|RD^*A\|^2 + a\|QA\|^2$; (3.129): $H_1B = G_1Q^*(QG_1Q^*)^{-1}B$. $G = (\Delta_\pi + DRD^* + Q^*aQ)^{-1}$ (3.122) p.420.

R2Algebra.lean:145–150,157–159; ZXR2Wire2.lean:61–62,77–78,86–87,90–91,71–72

```
noncomputable def Ginv : E →L[ℝ] E := D.Dpi + D.Ts.comp D.T + D.Mass
/-- `G` (3.122)/(3.27). -/
noncomputable def G : E →L[ℝ] E := Ring.inverse D.Ginv
/-- `H = G Q* (Q G Q*)⁻¹` (3.126). -/
noncomputable def H : F →L[ℝ] E :=
  (D.G.comp D.Qs).comp (Ring.inverse ((D.Q.comp D.G).comp D.Qs))
structure IsG1inv (G1inv : E →L[ℝ] E) : Prop where
  symm : ∀ x y, ⟪G1inv x, y⟫ = ⟪x, G1inv y⟫
  form : ∀ A, ⟪A, G1inv A⟫ = D.K A
noncomputable def G1inv : ↑(bondSupp P.G) →L[ℝ] ↑(bondSupp P.G) :=
  Classical.choose (P.data.exists_G1inv P.data_laws)
noncomputable def Qin : BondFnL2 (scaleUp P.ell P.k P.Lu) →L[ℝ] BondFnL2 P.Lu :=
  P.tau.comp P.sector.Qfull
noncomputable def H1in : BondFnL2 P.Lu →L[ℝ] BondFnL2 (scaleUp P.ell P.k P.Lu) :=
  (iE P.G).comp ((P.data.H1 P.G1inv).comp (ContinuousLinearMap.adjoint P.tau))
noncomputable def D2in (B : BondFnL2 P.Lu) : BondFnL2 P.Lu :=
  WithLp.toLp 2 (((P.ell : ℝ)⁻¹) · hLift P.Lam P.Q1 (C2tilde P.ell P.unitBgU (WithLp.ofLp B)))
noncomputable def Q1 : (Bond P.Lu → SuTwo) →₁[ℝ] (CoarseBond P.Lu P.ell → SuTwo) :=
  Q1jet P.ell P.unitBgU
```

In words. $G^{-1} = \Delta_\pi + DRD^* + Q^*aQ$ and G its inverse. G_1^{-1} is *the* symmetric operator whose quadratic form is the right side of (3.128) (the Lean choice is the unique such operator, proved to exist); G_1 its inverse. H_1, $\tilde{D}^{(2)}$ (a quadratic map built from the second-order term of the averaging, divided by ℓ), J and the unit-level Q_1 (the linear part of the one-step averaging, divided by ℓ) are built from the same ingredients; Q is the multi-level constraint operator followed by τ, which reads the top level.

Deviation. D10, D11 (jets, convention). The embedding τ and the placing of B at the top level are not in the print (A12). Not pinned: the identification of $\tilde{D}^{(2)} = hC^{(2)}$ with $h = L^{-1}\iota H_0^{-1}\rho$ (cited in the Lean comment to CMP109 p.267, a paper outside the corpus pages used here).

Check (yes/no). Does the chain $G^{-1} \to G \to G_1^{-1} \to G_1 \to H_1$ with QG_1Q^* inverted follow (3.122), (3.126)–(3.129)?

A11 The subspace, $\tilde{B}$ and the operator C

Print. CMP99b p.428 (p428:135): "This form is considered on the subspace $\{B : B = 0$ on $\Lambda^c,\ B = 0$ on $\bigcup_{y\in\Lambda'} Ax(y),\ Q_1B = 0\}\dots$ variables B restricted to the set of bonds $\tilde{\Lambda} = \Lambda\backslash(\bigcup_{y\in\Lambda'} Ax(y) \cup \bigcup_{c\in\Lambda'} B(c)\cap c)\dots$ $B = C\tilde{B}$, where C is a linear operator. It is an identity operator on almost all bonds, except the bonds b_0 for which a value $(C\tilde{B})(b_0)$ is equal to a solution of the equation $(QB)(c) = 0$." CMP96 p.245: "a set of bonds b such that at least one of the end-points b_-, b_+ belongs to Λ".

ZStatementP428.lean:211–213,216–218,230–233; ZR2A.lean:121–123; ZXR2Unit2.lean:98–101,104–107

```
def CRowIn {Lu : PeriodicLatticeExtent} (ell : ℕ) (Lam : Set (PeriodicLatticeSite Lu))
    (c : CoarseBond Lu ell) : Prop :=
  c.1.1 ∈ LamPrime ell Lam ∨ AdalyFormal.Balaban.CoarseCurl.coarseShift c.1.1 ell c.2 ∈ LamPrime ell Lam
def IsDroppedIn {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu)) (ell : ℕ)
    (b : Bond Lu) : Prop :=
  ∃ c : CoarseBond Lu ell, CRowIn ell Lam c ∧ b = droppedBondLast ell c
def OnLambdaTildeIn {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu)) (ell : ℕ)
    (B : Bond Lu → SuTwo) : Prop :=
  (∀ y ∈ LamPrime ell Lam, ∀ x ∈ blockOf y ell, ∀ b ∈ staircasePath y x, B b = 0) ∧
  (∀ b : Bond Lu, IsDroppedIn Lam ell b → B b = 0)
def OnLambdaT {Lu : PeriodicLatticeExtent} (Lam : Set (PeriodicLatticeSite Lu))
    (B : Bond Lu → SuTwo) : Prop :=
  ∀ b : Bond Lu, ¬ (b.1 ∈ Lam ∨ periodicLatticeShift b.1 b.2 ∈ Lam) → B b = 0
noncomputable def hLift {ell : ℕ} (Lam : Set (PeriodicLatticeSite L))
    (Q : (Bond L → SuTwo) →₁[ℝ] (CoarseBond L ell → SuTwo)) :
    (CoarseBond L ell → SuTwo) →₁[ℝ] (Bond L → SuTwo) :=
  iotaD ell Lam ∘₁ HOinv Lam Q ∘₁ rhoD ell Lam
noncomputable def Cop {ell : ℕ} (Lam : Set (PeriodicLatticeSite L))
    (Q : (Bond L → SuTwo) →₁[ℝ] (CoarseBond L ell → SuTwo)) :
    (Bond L → SuTwo) →₁[ℝ] (Bond L → SuTwo) :=
  LinearMap.id - hLift Lam Q ∘₁ Q
```

In words. $\tilde{B}$ is an $\mathfrak{su}(2)$-valued field on unit bonds that vanishes on the staircase paths $Ax(y)$ of each $y \in \Lambda'$, on each dropped bond $b_0(c)$ of a coarse bond c having an endpoint in Λ', and on every bond with no endpoint in Λ. $C = \mathrm{id} - hQ_1$ where h solves the constraint on the dropped bonds; it is the identity elsewhere. (The inverse of the constraint matrix H_0 is the junk 0 if H_0 is not bijective.)

Deviation. N1: "touching" (at least one endpoint) for $\tilde{B}$ and c, per CMP96 p.245; this is a larger set of $\tilde{B}$ than "$B = 0$ on Λ^c" with both endpoints (Lean `InstIneq.toBoth`: the touching form implies the both-endpoint form).

Check (yes/no). Is $\tilde{B}$ (vanishing on $Ax(y)$ and on the dropped bonds b_0, with the "at least one endpoint" bond convention) the variable set of p.428, and is $C = \mathrm{id} - hQ_1$ "the operator that solves $(QB)(c) = 0$ on b_0"?

A12 The form, the norm, the inequality

Print. CMP99b (3.156) p.428: $\langle B, (QG_1Q^*)^{-1}B\rangle - a\langle B, B\rangle - 2\langle H_1\tilde{D}^{(2)}(B), J\rangle = \langle B, \Delta_k B\rangle$. Norm: p.390 (p390:129): "a norm $|X|$ of a $N \times N$ matrix means the Hilbert–Schmidt norm: $|X|^2 = \mathrm{tr}X^*X$"; p.392 (p392:161): "the trace is normalized, i.e., $\mathrm{tr}\,1 = 1$".

ZXG3A.lean:50–51,54–55,58–58,63–66; RegularUPerturbation.lean:355–356

```
noncomputable def firstTerm (B : BondFnL2 P.Lu) : ℝ :=
  P.eta * ⟪ContinuousLinearMap.adjoint P.tau B, Ring.inverse (Nmat P) (ContinuousLinearMap.adjoint P.tau B)⟫
noncomputable def genForm (B : BondFnL2 P.Lu) : ℝ :=
  firstTerm P B - P.a * ⟪B, B⟫ - 2 * ⟪P.H1in (P.D2in B), P.Jf⟫
noncomputable def genC : (Bond P.Lu → SuTwo) →₁[ℝ] (Bond P.Lu → SuTwo) := Cop P.Lam P.Q1
abbrev gparams (ell k : ℕ) (Lu : PeriodicLatticeExtent) (hell : 0 < ell)
    (S : Tower (scaleUp ell k Lu)) (U : LinkCfg (scaleUp ell k Lu)) (a : ℝ)
    (Lam : Set (PeriodicLatticeSite Lu)) : IngrBParams :=
  ⟨ell, k, Lu, hell, S, U, a, Lam⟩
noncomputable def bondEnergy (A : Bond L → SuTwo) : ℝ :=
  ∑ b : Bond L, ‖A b‖ ^ 2
```

In words. `genForm` is the form of (3.156), with B placed at the top level by τ^* and the first term multiplied by η; the bilinear pairing of B with itself is the unit-bond L^2 pairing. `genC` is C. `bondEnergy` $\tilde{B} \mapsto \sum_b \|\tilde{B}(b)\|^2$, with the norm on $\mathfrak{su}(2)$ equal to the printed trace-normalised Hilbert–Schmidt norm (Lean `norm_sq_eq_half_frobSq`). The inverse is Lean's: where QG_1Q^* is not invertible the first term is 0.

Deviation. Not in the table: the factor η and τ^* in the first term are a normalisation of the formalisation (paper §3.2 states there is no deviation row for them and that the inner-product weights are inherited, not re-derived). They are not pinned to a printed page. The junk value 0 makes the statement stronger than one that presupposes existence of the inverses (paper Remark 3.2).

Check (yes/no). Is $\|\tilde{B}\|^2 = \sum_b \|\tilde{B}(b)\|^2$ with trace-normalised norm the norm of the printed lower bound $\gamma_0\|\tilde{B}\|^2$?

A13 The inequality, the inner claim, the constants, the quantifier order

Print. CMP99b p.428 (p428:153): "It is more convenient to work with the operator $\tilde{C}^{(k)}(\Lambda)$, because it is defined by a positive definite operator $C^*\Delta_k C$ with a lower bound $\gamma_0 > 0$ independent of k and U. We have proved it in [4], Lemma 2.4, for operators with $U = 1$. Localizing the operators in Δ_k and using the methods of Sect. B we can prove it for $C^*\Delta_k C$ with an arbitrary configuration U satisfying (3.35), (3.36) with $M\alpha_0$ sufficiently small." R, M: p.408 (p408:131): "We take these numbers as powers of L."

ZT5A.lean:56–61,89–96,170–173,200–201,229–230,291–294

```
def InstIneq (ell k : ℕ) (hell : 0 < ell) (a : ℝ) (Lu : PeriodicLatticeExtent) (γ₀ : ℝ)
    (S : Tower (scaleUp ell k Lu)) (Lam : Set (PeriodicLatticeSite Lu)) (U : LinkCfg (scaleUp ell k Lu)) : Prop :=
  ∀ (Bt : Bond Lu → SuTwo), OnLambdaTildeIn Lam ell Bt → OnLambdaT Lam Bt →
    γ₀ * bondEnergy Bt ≤
      GenP.genForm (gparams ell k Lu hell S U a Lam)
        (WithLp.toLp 2 (GenP.genC (gparams ell k Lu hell S U a Lam) Bt))
def CoercG8At (ell M R k : ℕ) (hell : 0 < ell) (a : ℝ) (Lu : PeriodicLatticeExtent)
    (c : ℕ) (α₀ γ₀ : ℝ) : Prop :=
  TorusOK ell M Lu →
  ∀ (S : Tower (scaleUp ell k Lu)), Admissible ell M R k S →
  ∀ (Lam : Set (PeriodicLatticeSite Lu)), LambdaOK2 ell M R k S Lam →
  ∀ (U : LinkCfg (scaleUp ell k Lu)),
    ClassCoupled ell M c k S α₀ ((ell : ℝ) ^ k)⁻¹ U →
    InstIneq ell k hell a Lu γ₀ S Lam U
def P428Core8At (ell M R : ℕ) (hell : 0 < ell) (a : ℝ) (c : ℕ) : Prop :=
  ∃ a₀ : ℝ, 0 < a₀ ∧ ∃ γ₀ : ℝ, 0 < γ₀ ∧
    ∀ (k : ℕ), 1 ≤ k → ∀ (Lu : PeriodicLatticeExtent) (α₀ : ℝ), 0 < α₀ → (c : ℝ) * M * α₀ ≤ a₀ →
      CoercG8At ell M R k hell a Lu c α₀ γ₀
def SidePow6 (ell M : ℕ) (a : ℝ) : Prop :=
  2 ≤ ell ∧ (∃ m : ℕ, 1 ≤ m ∧ M = ell ^ m) ∧ 0 < a
def P428Genuine8 (ell M R : ℕ) (hell : 0 < ell) (a : ℝ) : Prop :=
  SidePow6 ell M a ∧ P428Core8 ell M R hell a
def P428Target8 : Prop :=
  ∀ (ell : ℕ) (hell : 2 ≤ ell) (a : ℝ), 0 < a →
    ∃ m₀ : ℕ, 1 ≤ m₀ ∧ ∀ m : ℕ, m₀ ≤ m → ∃ R₀ : ℕ, ∀ R : ℕ, R₀ ≤ R →
      P428Genuine8 ell (ell ^ m) R (by omega) a
```

In words. `InstIneq`: for every admissible $\tilde{B}$, $\gamma_0\|\tilde{B}\|^2 \le \mathrm{genForm}(C\tilde{B})$. `CoercG8At`: for tori, towers, regions Λ and configurations U satisfying all the hypotheses above, `InstIneq` holds. `P428Core8At`: there are $a_0, \gamma_0 > 0$ (chosen *before* k) such that for every $k \ge 1$, torus, and $\alpha_0 > 0$ with $cM\alpha_0 \le a_0$, `CoercG8At` holds; the headline has $c = 12$. `P428Target8`: for every $\ell \ge 2$ and $a > 0$ there is m_0 such that for all $m \ge m_0$ there is R_0 such that for all $R \ge R_0$ (with $M = \ell^m$) the above holds.

Deviation. D4: a_0, γ_0 are independent of $k \ge 1$, torus, α_0, tower, Λ, U, $\tilde{B}$ (the print: "independent of k and U"); they may depend on ℓ, a, M, R. D5, D6: all $a > 0$ and all $\ell \ge 2$. D8: $k \ge 1$. N5: $M = \ell^m$. "$M\alpha_0$ sufficiently small" is $12M\alpha_0 \le a_0$ with a_0 existential. "M, R sufficiently large" is $m \ge m_0$, $R \ge R_0$, both existential.

Check (yes/no). Is the prover's order of choice (the constants a_0, γ_0 fixed after ℓ, a, m, R and before k, torus, α_0, tower, Λ, U, $\tilde{B}$) what "independent of k and U" means?

Part B. The whole statement in ordinary notation

Write $M = \ell^m$, $\eta = \ell^{-k}$, $N_\mu = \mathrm{Lu}_\mu + 1$ (unit sites per side), fine torus with $N_\mu \ell^k$ sites per side. **Who chooses:** $\forall$ = the adversary, $\exists$ = the prover; the order is left to right.

$$
\begin{aligned}
&\forall \ell \ge 2\ \forall a > 0\ \ \exists m_0 \ge 1\ \ \forall m \ge m_0\ \ \exists R_0\ \ \forall R \ge R_0 : \\
&\quad \exists a_0 > 0\ \ \exists \gamma_0 > 0\ \ \forall k \ge 1\ \ \forall \mathrm{Lu}\ \ \forall \alpha_0 > 0 \text{ with } 12M\alpha_0 \le a_0 : \\
&\qquad \text{if } \ell \mid N_\mu,\ \ell < N_\mu,\ M \mid N_\mu\ (\mu = 1,2,3),\ \text{then} \\
&\qquad \forall \text{ towers } S = (\Omega_j)_{j \ge 0} \text{ on the fine torus with } \mathrm{Admissible}(\ell, M, R, k, S) \\
&\qquad \forall \Lambda \subset \text{unit torus with } \mathrm{LambdaOK2}(\ell, M, R, k, S, \Lambda) \\
&\qquad \forall U : \text{fine bonds} \to SU(2) \text{ with } \mathrm{ClassCoupled}(\ell, M, 12, k, S, \alpha_0, \eta, U) : \\
&\qquad\quad \forall \tilde{B} : \text{unit bonds} \to \mathfrak{su}(2) \text{ admissible (A11)} : \\
&\qquad\qquad \gamma_0 \sum_b \|\tilde{B}(b)\|^2 \ \le\ \mathrm{genForm}_{(S,U,a,\Lambda)}(C\tilde{B}).
\end{aligned}
$$

where $\mathrm{genForm}(B) = \eta\langle \tau^* B, (QG_1Q^*)^{-1}\tau^* B\rangle - a\langle B, B\rangle - 2\langle H_1 \tilde{D}^{(2)}(B), J\rangle$ with all operators built from (S, U, a, Λ) as in A8–A12. $\mathrm{ClassCoupled}(c)$ = [every site and every Ω_0-plaquette lies in a class cube of factor c] $\wedge$ [$\forall$ class cube $\square$ of factor c and index j: $\exists g, A$ with $U^g = e^{i\eta A}$ on $\square$ and (3.35), (3.36) with $cM\alpha_0$]. S ranges over all admissible towers, not over one chosen by the prover; γ_0 does not depend on k, U, S, Λ, $\tilde{B}$, α_0 or the torus. Consequences visible in the text: (i) no hypothesis involves a_0 except $12M\alpha_0 \le a_0$, which is satisfiable for any a_0 by taking α_0 small, so the prover cannot empty the statement by its choice of a_0; (ii) the thresholds m_0, R_0 are existential and unnumbered; (iii) γ_0 may depend on ℓ, a, m, R (the print names neither M nor R).

Part C. Stated readings: what changes if rejected, and what is proved instead

Lean names are in the appendix with file and line. "Proved" means a Lean theorem in the project's rebuilt tree, depending only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`.

#	reading	if the reader rejects it	proved alternative (Lean)
1	**Reading A of** T_1 (p.396): T_1 is the fine lattice in its own unit; blocks of Ω_0 have M fine sites, radius RM fine sites.	Reading B (T_1 = level-k unit lattice) gives a larger Ω_0 built from blocks of $M\ell^k$ fine sites. The frozen statement itself covers only A.	`target8Layer` (`Target8Layer`, `THL6.lean:164`) drops the clause (6) and keeps only the strict collar and block closure, so any block-closed Ω_0 with the collar is covered. The audit theorems `readingB_satisfies_collar`, `readingB_fine_blockClosed` (file `sol_review_r_bridge_20261019/work/Audit.lean`, outside the frozen modules) say reading B's set has these two properties; not re-derived here. The identity of `Admissible` with the printed tower under A is `RBridge.admissible_iff_print`.
2	**Choice of** Ω_0 **(p.396/p.394)**: the layer $\leq RM$ ("e.g."); a thinner layer is permitted by p.394.	A thinner (smaller) block-closed Ω_0 is not covered by the frozen statement. A layer must still satisfy the strict collar R_0M, $R_0 \geq R_1$.	`target8Layer`: `P428Target8` with `Admissible` replaced by `AdmNoBase` (no clause 6; one radius). Class cubes read from the same Ω_0. Two-domain form (operators on a thin Ω_0^{op}, class on a larger $\Omega_0^{cl} \supseteq \Omega_0^{op}$): `target8TwoDom` (`TD4.lean:74`); for unions, boxes, level-0 caps: `famTwoDom_union`, `famTwoDom_box`, `famTwoDom_cap` (`TD2.lean:181`, `TD3.lean:98,90`). *Not covered*: collars below the sufficient threshold R_1 (existential, no value).
3	**One cube factor** $c = 12$, one factor for the whole class (p.409).	The print attaches $O(1) \geq 10$ to each cube.	`P428Genuine8AllC` (every $c \geq 10$ fixed first; `ZT5A.lean:257`); `P428Target8Total` (`TT_F.lean:29`: constants fixed before $c \geq 10$); `P428Target8FamilyTorus` (`MXQ_Tor.lean:58`): each cube its own factor $f \geq 10$.
4	**Cubes are boxes** of side $fM\ell^j$.	The print says "union of several big blocks".	`P428Target8FamilyTorusU` (`UNI_Tor.lean:73`), with `IsClassUnion`: arbitrary unions of grid cells; implies the box statement. Level-0 cap class (cubes may leave Ω_0; regularity on cube $\cap\,\Omega_0$): `FamTorusCapLayer` (`THL6.lean:99`), `ClassCoupledCap` (`LOC2.lean:136`). The factor is a label not the size, and the block scale is $M\ell^j$ fine sites (not the literal subscript $T_{L^{-j}}$ with $\eta = \ell^{-k}$).
5	**Bond convention**: class bonds have *both* endpoints in the cube (`CubeBond`); $\tilde{B}$ uses "at least one" (N1). Print: CMP96 p.224 says at least one for domains; CMP102 p.277 states it as a general convention: domains are identified "as sets of bonds with at least one end-point belonging to Ω_j, or sets of plaquettes with at least one corner belonging to Ω_j. This remark applies to other sets also."	For the class: if class bonds must be touching bonds, U is constrained on more bonds, so the hypothesis on U is stronger and the claim is weaker; the frozen "both endpoints" class is the larger class (expected to imply the touching version, not yet a Lean theorem, since the coverage clause also uses `CubeBond`). For the operators: if the domains Ω_j, Λ themselves are read as touching bond and corner plaquette sets, as CMP102 p.277 says, the operators change; see row 7 (plaquette selection).	None proved. Lean theorem `boundary_bond_omitted` (audit file, outside the frozen modules) shows, on a one-site domain, that `CubeBond` omits a bond incident to it; no theorem converts "both" to "at least one".
6	**Coverage** (A7) is an added hypothesis.	If dropped, the encoded statement is *false*: `P428Target8NC` (coverage dropped) is refuted at $\ell = 3$, $a = \frac{1}{2}$, via an in-class U at which G^{-1} is not invertible (Lean junk value 0). Whether a coverage-free lemma presupposing the printed inverses holds is not decided.	`coverage_necessary` (`P428Target8` $\wedge \neg$`P428Target8NC`, `CvN7.lean:15`), `tip_counterexample` (`CvN6.lean:144`), `not_P428Target8NC` (`CvN6.lean:183`). No variant without coverage.

#	reading	if the reader rejects it	proved alternative (Lean)
7	**Plaquette selection** (A9): Δ, J over plaquettes inside Ω_0 (`plaqIn`).	The print sums (3.10), (3.11) over all plaquettes. Literal readings are refuted (the Lean statement fails where G^{-1} is not a unit), but whether the printed claim fails where all inverses exist is not decided.	`P428Target8S selIn selIn = P428Target8` (`rfl`, `D1Lit.lean:134`); `not_P428Target8Literal` (all plaquettes, `D1Final.lean:169`); `not_P428Target8LiteralBoth` (`D2Final.lean:88`). No proved variant for the all-plaquette reading.
8	**Bond/norm and normalisations**: Frobenius norm in the class ($\sqrt{2}$, $\sqrt{2}$); η and τ^* in the first term of `genForm`; jets of Q (D10, D11).	Constants change; a_0 absorbs the norm factor. The η-factor and τ^* are not checked against print.	None.
9	**Junk-value inverse** (0 where non-invertible).	A version that presupposes the inverses exist is a different statement; the frozen one is stronger.	`p428Target8Total` proves the form with existence of inverses in the conclusion (`CoercG8AtWD`); `TT_F.lean:29`.

Appendix. Sources (file:line)

Frozen statement: `compute/statement_lane_20261012_r8/fork/ZT5A.lean` (`P428Target8` at 291; `CoercG8At` 89; `InstIneq` 56; `P428Core8At` 170; `SidePow6` 200; `P428Genuine8` 229); proof: `AdalyFormal.M2RightInv.p428Target8`, `compute/m2_rightinv_20261015/src/RI6.lean:323`. Tower and class: `ZStatementP428.lean`; Λ: `ZT4A.lean`; regularity: `ZR2A.lean`; coverage: `ZR6A.lean`; operators: `ZR2Ops2.lean`, `ZXR2Wire1.lean`, `ZXR2Wire2.lean`, `ZXR2Unit2.lean`, `ZXG3A.lean`; base operators: `compute/ingredient_family_20261009_r2/R2Ops1.lean`, `R2Algebra.lean`; norm: `compute/forall_u_rung/deps/AdalyFormal/Balaban/RegularUPerturbation.lean:355`. Variants: `compute/thin_layer_20261019/src/THL6.lean`, `THL2.lean:23`; `compute/two_domain_20261019/src/TD2-TD4.lean`; `compute/union_cubes_20261018/src/UNI_Tor.lean`; `compute/total_20261017/src/TT_F.lean`; `compute/mixed_20261017/src_q/MXQ_Tor.lean`; `compute/level0_class_20261019/src/L0C2.lean`; `compute/cover_necessary_20261016/src/CvN6-7.lean`; `compute/d1_literal_20261018/src/D1Lit.lean`; `compute/r_bridge_20261019/src/RB1.lean`. Print: `corpus/pages/CMP99b_p{390,392,393,394,396,408,409,419,420,421,427,428}.md`, `CMP96_p{223,224,245}.md`; images `corpus/renders/CMP99b_p396-08.png` (p.396 quotes read at the image) and, for pp.393–394, 408, 427, `compute/r_bridge_20261019/images/`. Extraction commands and sha256 of every source file: `CHECKS.md`.